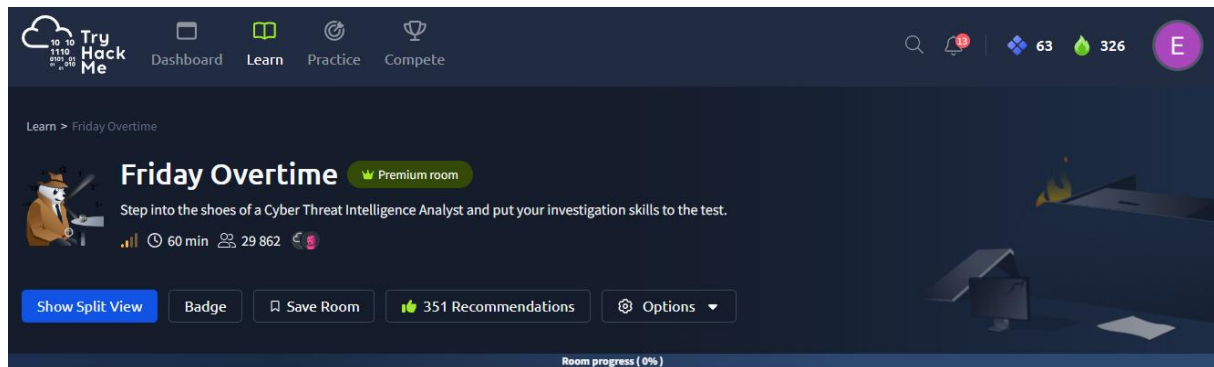


Introduction



Hello Busy Weekend. . .

It's a Friday evening at PandaProbe Intelligence when a notification appears on your CTI platform. While most are already looking forward to the weekend, you realise you must pull overtime because SwiftSpend Finance has opened a new ticket, raising concerns about potential malware threats. The finance company, known for its meticulous security measures, stumbled upon something suspicious and wanted immediate expert analysis.

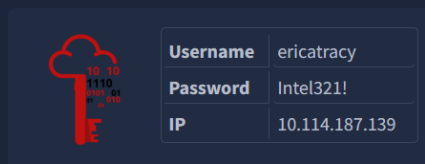
As the only remaining CTI Analyst on shift at PandaProbe Intelligence, you quickly took charge of the situation, realising the gravity of a potential breach at a financial institution. The ticket contained multiple file attachments, presumed to be malware samples.

With a deep breath, a focused mind, and the longing desire to go home, you began the process of:

1. Downloading the malware samples provided in the ticket, ensuring they were contained in a secure environment.
2. Running the samples through preliminary automated malware analysis tools to get a quick overview.
3. Deep diving into a manual analysis, understanding the malware's behaviour, and identifying its communication patterns.
4. Correlating findings with global threat intelligence databases to identify known signatures or behaviours.
5. Compiling a comprehensive report with mitigation and recovery steps, ensuring SwiftSpend Finance could swiftly address potential threats.

Connecting to the machine

Start the lab machine in split-screen view by clicking the green **Start Lab Machine** button on the upper right section of this task. If the VM is not visible, use the blue **Show Split View** button at the top-right of the page. Additionally, you can open the DocIntel platform using the credentials below.



Note: While the web browser (i.e., Chromium) will immediately start after boot up, it may show a tab that has a "502 Bad Gateway" error message displayed. This is because the DocIntel platform takes about 5 more minutes to finish starting up **after** the VM has completely booted up. After 5 minutes, you can **refresh** the page in order to view the login page. We appreciate your patience. The ticket details can be found by logging in to the DocIntel platform. **OSINT, a web browser, and a text editor outside the VM will also help.**

Hello everyone, today we've become CTI Analyst for CTI company, which offers analysis of the malware that companies had dealt with already to better understand what exactly was going on during the attack.

Investigation of incident

This is the email address from Oliver Bennett, who works in the cybersecurity department at SwiftSpend Finance company, which found out a problem and intrusion in its system. In the attached zip. file we were given a sample of the malware that was found in the company's system. So, we are going through it to find out what the malware is made of.



Urgent: Malicious Malware Artefacts Detected
by **SwiftSpend Finance** - Registered 2023-12-07T23:10:51.5503900
Dear PandaProbe Intel team,

I hope this message finds you well. My name is Oliver Bennett from the Cybersecurity Division at SwiftSpend Finance. During our recent security sweep, we have identified a set of malicious files which, based on our preliminary analysis, seem to be associated with .

Details

Date Detected: Friday, December 8, 2023

Infected Systems: Over 9000 systems

Nature of Malware: Unknown / Suspected RAT

We believe the intent of this malware is to gain a foothold to ultimately exfiltrate sensitive financial data and possibly deploy ransomware.

Immediate Actions Taken

- Isolated the infected systems from the network.
- Initiated a comprehensive scan across all systems.
- Collected and stored malware samples securely for further analysis.
- We are currently collaborating with external cybersecurity agencies and our security solutions providers to get a deeper understanding of this malware. However, we wanted to raise this with you immediately given the potential risk associated with APTs.

We strongly need your team's assistance with conducting a thorough review of the malware sample. The password to the attached archive is: **Panda321!**

Moving forward, we are going to conduct a User Awareness Training to inform all staff members to be extra cautious, especially when dealing with email attachments and links.

Attached are the indicators of compromise (IoCs) for your perusal. I am also available for a call or meeting to discuss our findings in detail and strategise our response.

Your prompt attention to this matter is highly appreciated. Let's work together to ensure the safety and integrity of our systems and data.

Warm regards,
Oliver Bennett
Cybersecurity Division
SwiftSpend Finance
Phone: +123 456 7890
Email: oliver.bennett@swiftspend.finance

Files Attachments [View all](#)



[samples.zip](#)
2023-12-07T23:10:51.8621190

Before we begin with the opening of the sample. We always need to pay attention to our surroundings, it's still malware after all, so dynamic analysis must be done in safe environment, for example: Sandbox VM, which doesn't affect our workstation.

At first, we unzipped (**unzip samples.zip**) the file and put the password in: **Panda321!**

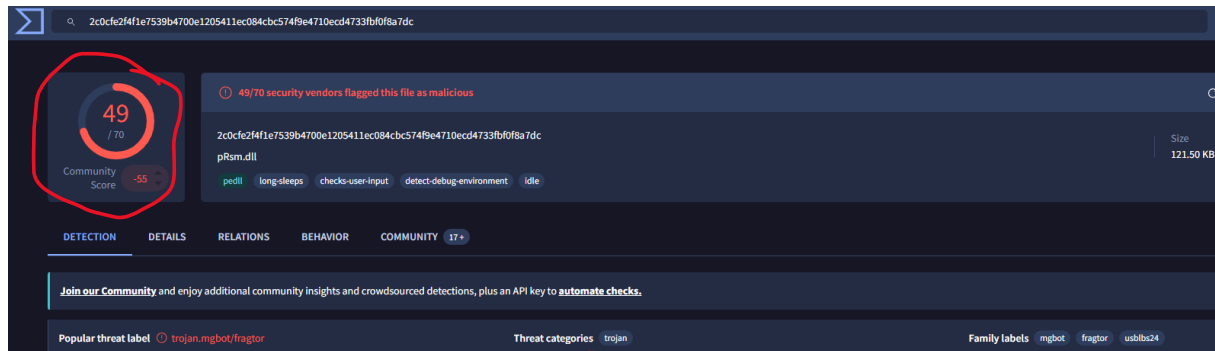
```
Impacting: webserver.dll  
[ericatracy@ip-10-114-187-139 Downloads]$ ls  
cbmrpa.dll maillfpassword.dll pRsm.dll qmsdp.dll samples.zip wdbcrc.dll  
[ericatracy@ip-10-114-187-139 Downloads]$
```

These are all the files contained in the zip file.

Following the lab questions, we created a SHA1 hash of the pRsm.dll file:

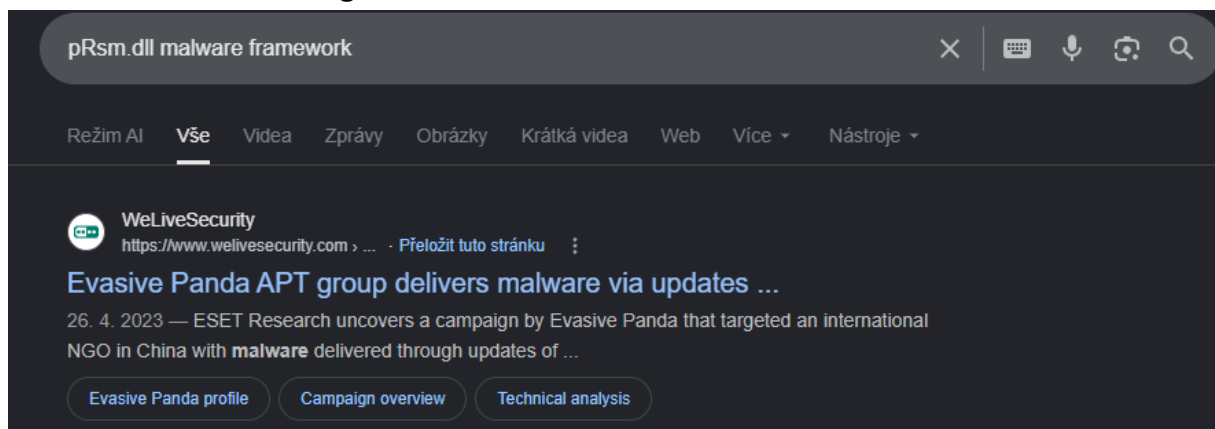
`sha1sum pRsm.dll` > **9d1ecbbe8637fed0d89fca1af35ea821277ad2e8**

Then we put the hash into CTI database > VirusTotal or Malware Bazaar



From VirusTotal database we can see that the .dll add-on is clearly malicious and is part of mgbot family, which is trojan category level threat.

Next, we had to find out which malware framework utilizes these DLLs as add-on modules, which we have already done, but the purpose of this lab was also to use OSINT. So, we used Google for it:



And found out ESET article from 2023, which helped us understand what is specifically going on and how the malware works.

Article: <https://www.welivesecurity.com/2023/04/26/evasive-panda-apt-group-malware-updates-popular-chinese-software/#technical-analysis>

Toolset

MgBot

MgBot is the primary Windows backdoor used by Evasive Panda, which according to our findings has existed since at least 2012 and, as mentioned in this blog post, was publicly [documented at VirusBulletin in 2014](#). It was developed in C++ with an object-oriented design, and has the capabilities to communicate via TCP and UDP, and extend its functionality via plugin modules.

MgBot's installer and backdoor, and their functionality, have not changed significantly since it was first documented. Its chain of execution is the same as described in this [report](#) by Malwarebytes from 2020.

MgBot Plugins

MgBot's modular architecture allows it to extend its functionality by receiving and deploying modules on the compromised machine. Table 2 lists the known plugins and their functionality. It is important to note that the plugins don't have unique internal identification numbers; therefore we are identifying them here by their DLL names on disk, which we have never seen change.

pRsm.dll	Captures input and output audio streams.
----------	--

To the extension file – pRsm.dll is linked MITRE ATTACK technique – T1123 (MgBot's plugin module pRsm.dll captures input and output audio streams).

Malware was first seen on the 2nd November 2020 and was downloaded from:
hxxp[:]//[update[.]browser[.]qq[.]com/qmbs/QQ/qqUrlMgr_QQ88_4296.exe

For safer usage we defanged it in CyberChef.

Defanged IP address of the C&C server first detected on 2020-09-14 using these modules can yet again be found in the same article:

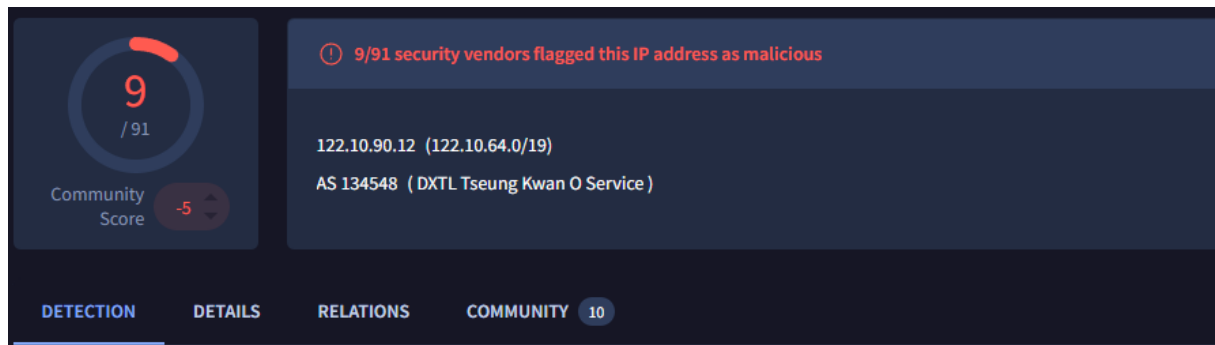
Network

IP	Provider	First seen	Details
122.10.88[.]226	AS55933 Cloudie Limited	2020-07-09	MgBot C&C server.
122.10.90[.]12	AS55933 Cloudie Limited	2020-09-14	MgBot C&C server.

Defanged version of the IP address: **122[.]10[.]90[.]12**

For the last task we had to find out md5 hash of the spyagent family spyware hosted on the same IP targeting Android devices in June 2025.

We know that it used the same IP so we will search in Virus Total for more information

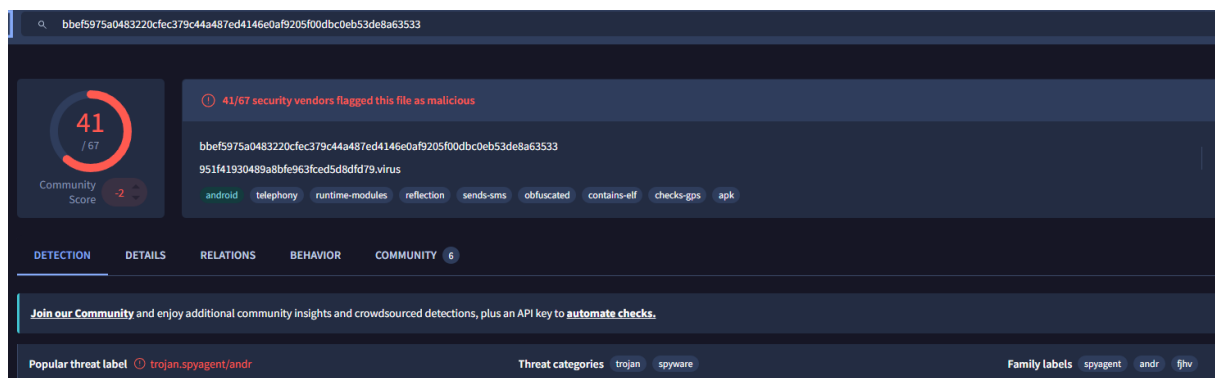


The screenshot shows the VirusTotal interface for an IP address. On the left, a circular progress indicator shows a score of 9 out of 91, with a community score of -5. The main header indicates that 9/91 security vendors flagged this IP address as malicious. Below this, the IP address 122.10.90.12 (122.10.64.0/19) is listed, along with its AS number, AS 134548 (DXTL Tseung Kwan O Service). At the bottom, there are tabs for DETECTION, DETAILS, RELATIONS, and COMMUNITY (10).

Scanned	Detections	Type	Name
2025-03-13	60 / 73	Win32 EXE	flashplayerax_install.exe
2025-10-10	53 / 72	Win32 EXE	ald_j.exe
2024-08-10	56 / 75	Win32 EXE	flashplayer_install_cn.exe
2025-12-02	41 / 67	Android	951F41930489A8BFE963FCED5D8DFD79

Communicating Files (4)			
Scanned	Detections	Type	Name
2025-03-13	60 / 73	Win32 EXE	flashplayerax_install.exe
2025-10-10	53 / 72	Win32 EXE	ald_j.exe
2024-08-10	56 / 75	Win32 EXE	flashplayer_install_cn.exe
2025-12-02	41 / 67	Android	951F41930489A8BFE963FCED5D8DFD79

After getting through each of these malicious files, the only one with spyware related activities was the last one: 951f41930489a8bfe963fced5d8dfd79.virus



The screenshot shows the VirusTotal interface for a file. The search bar contains the MD5 hash: bbf5975a0483220cfec379c44a487ed4146e0af9205f00dbc0eb53de8a63533. On the left, a circular progress indicator shows a score of 41 out of 67, with a community score of -2. The main header indicates that 41/67 security vendors flagged this file as malicious. Below this, the file name is listed: bbf5975a0483220cfec379c44a487ed4146e0af9205f00dbc0eb53de8a63533 951f41930489a8bfe963fced5d8dfd79.virus. A row of tags indicates the file's characteristics: android, telephony, runtime-modules, reflection, sends-sms, obfuscated, contains-elf, checks-gps, and apk. At the bottom, there are tabs for DETECTION, DETAILS, RELATIONS, BEHAVIOR, and COMMUNITY (6). A banner at the bottom encourages joining the community. The bottom status bar shows a popular threat label: trojan.spyagent/andr, threat categories: trojan, spyware, and family labels: spyagent, andr, fjhv.

MD5 hash of this file is : 951f41930489a8bfe963fced5d8dfd79

(<https://www.virustotal.com/gui/file/bbf5975a0483220cfec379c44a487ed4146e0af9205f00dbc0eb53de8a63533/details>)

FINAL WORLDS

The compromise at SwiftSpend Finance must be treated as a highly targeted cyber espionage incident rather than a baseline malware infection. The deployment of specific audio-surveillance plugins indicates an intent to intercept sensitive corporate discussions, partner negotiations, or internal communications.

Network defenders must immediately block all associated Indicators of Compromise (IoCs) and audit systems for unauthorized audio device interactions.